

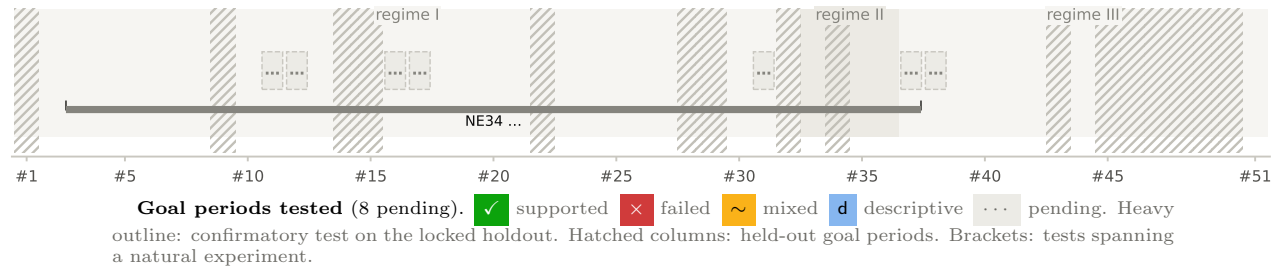
H10: Goals are Legendre pushes: a swarm’s response to an assigned goal is predictable from its unforced fluctuations

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

not yet rated

Bottom line. Pending.

level: hypothesis · pipeline checks 57% · updated —



WHAT HAPPENED

PHYSICAL MODEL

From: physics-models/11-vector-spins (mean-field $O(n)$: $m = L_n(\beta(J_0 - m + h))$); susceptibility $\chi = N \cdot \text{var}(m)$), with physics-models/01-inverse-ising for the scalar Curie–Weiss analogue. One field parameter h per assigned week is fitted to the mean shift; the variance and higher cumulants under the field are then *predicted* (an unfitted test, axis D).

Not yet summarized. The question: Estimate the free energy $F(m)$ of the swarm’s alignment m from its fluctuations in field-free (pick-your-own-goal) weeks. Can it predict how the swarm lines up behind an assigned goal, by tilting $G = F - h \cdot m$, without refitting? Practical payoff: forecasting how steerable a swarm is from how it behaves when nobody steers it.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Observables. Pending.

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